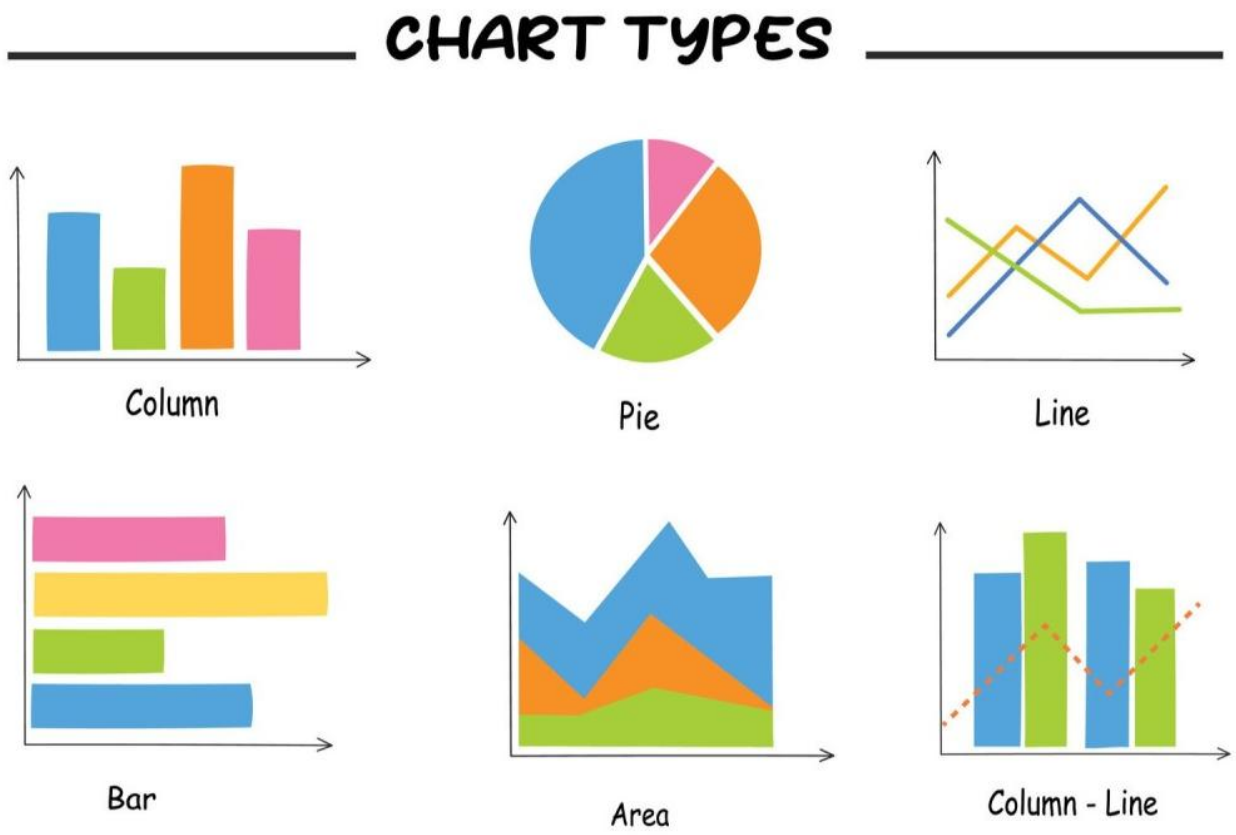
Session 15: Visualizing Data with Charts

Presented by

Ravindu Sooriyaarachchi

Session objectives

- *Learn to insert and customize Bar, Line, and Pie Charts in Excel.*
- *Understand when to use each type of chart.*
- *Practice with real examples.*



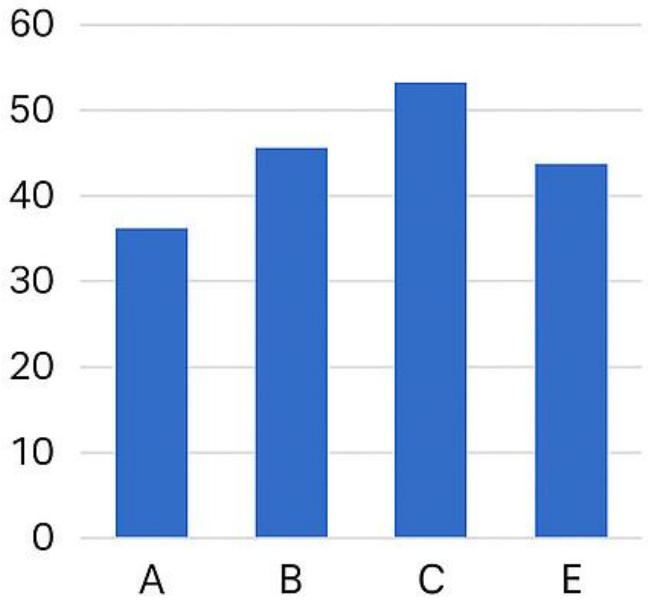
Why Use Charts?

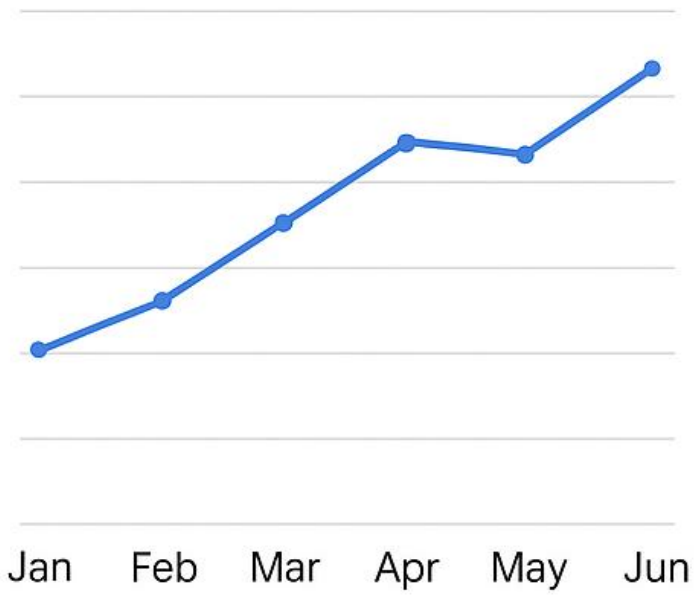
Table vs. Chart: Which One Helps You Make Better Decisions?

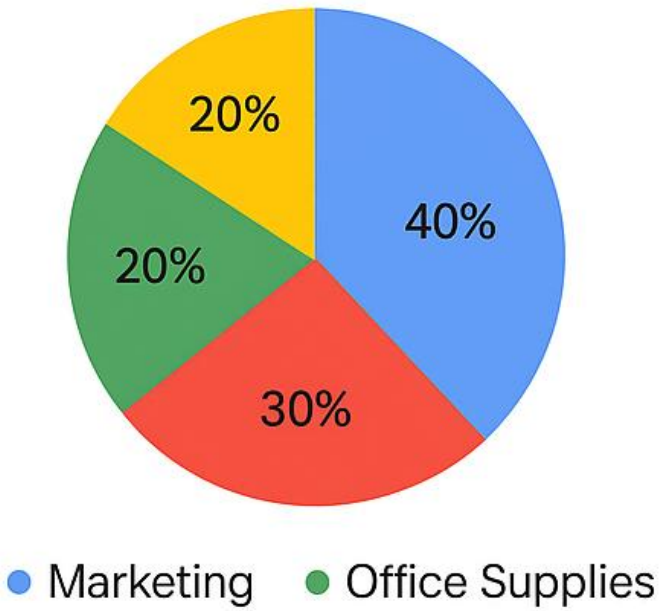
- *Charts help communicate data clearly and quickly.*
- *Makes trends and comparisons easy to see.*
- *Ideal for presentations and reports.*

Types of Charts

- *Bar/Column Chart – Compare categories*
- *Line Chart – Show trends over time*
- *Pie Chart – Show proportions of a whole*

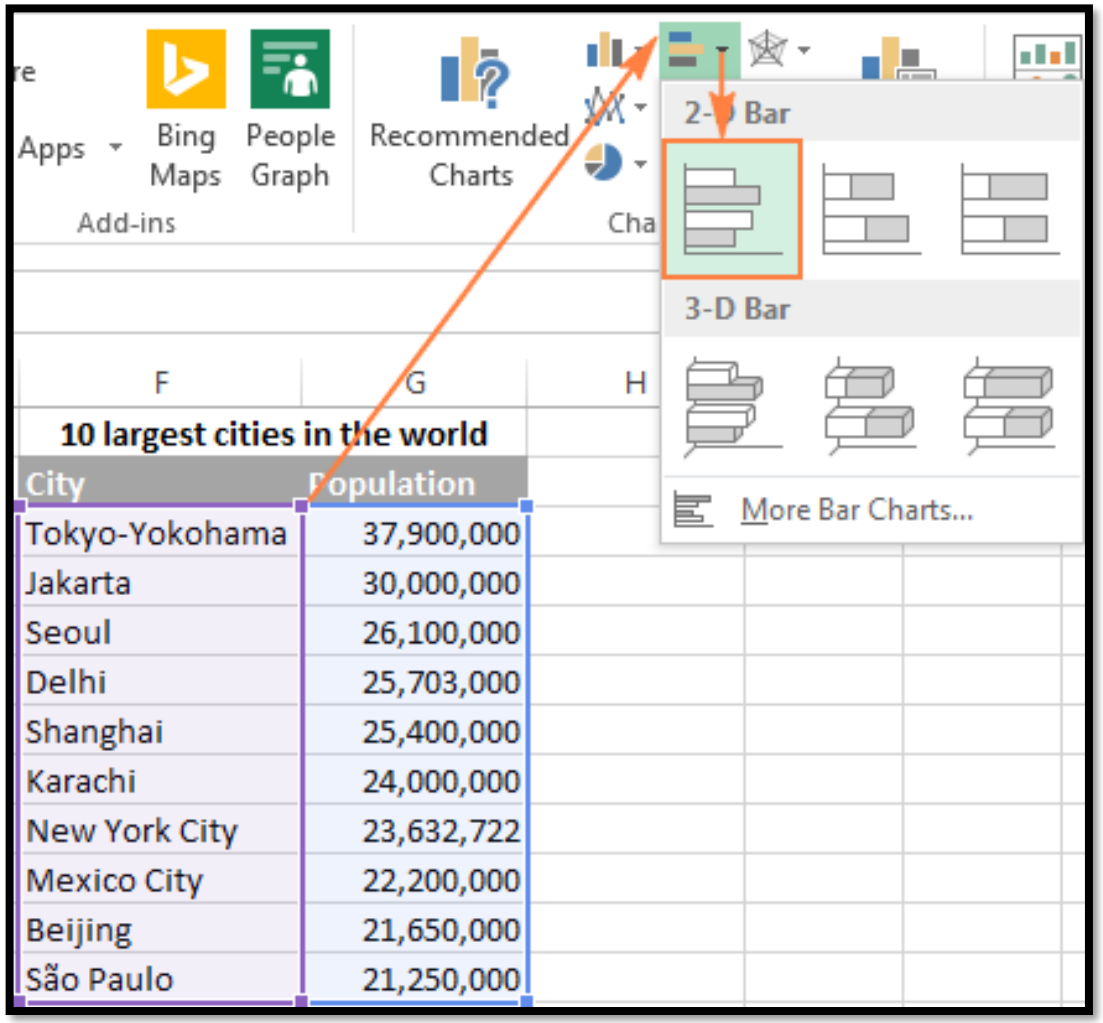






Creating a Bar Chart

1. *Select your data*
2. *Go to Insert → Chart → Column or Bar*
3. *Choose a style*
4. *Customize using Chart Design*

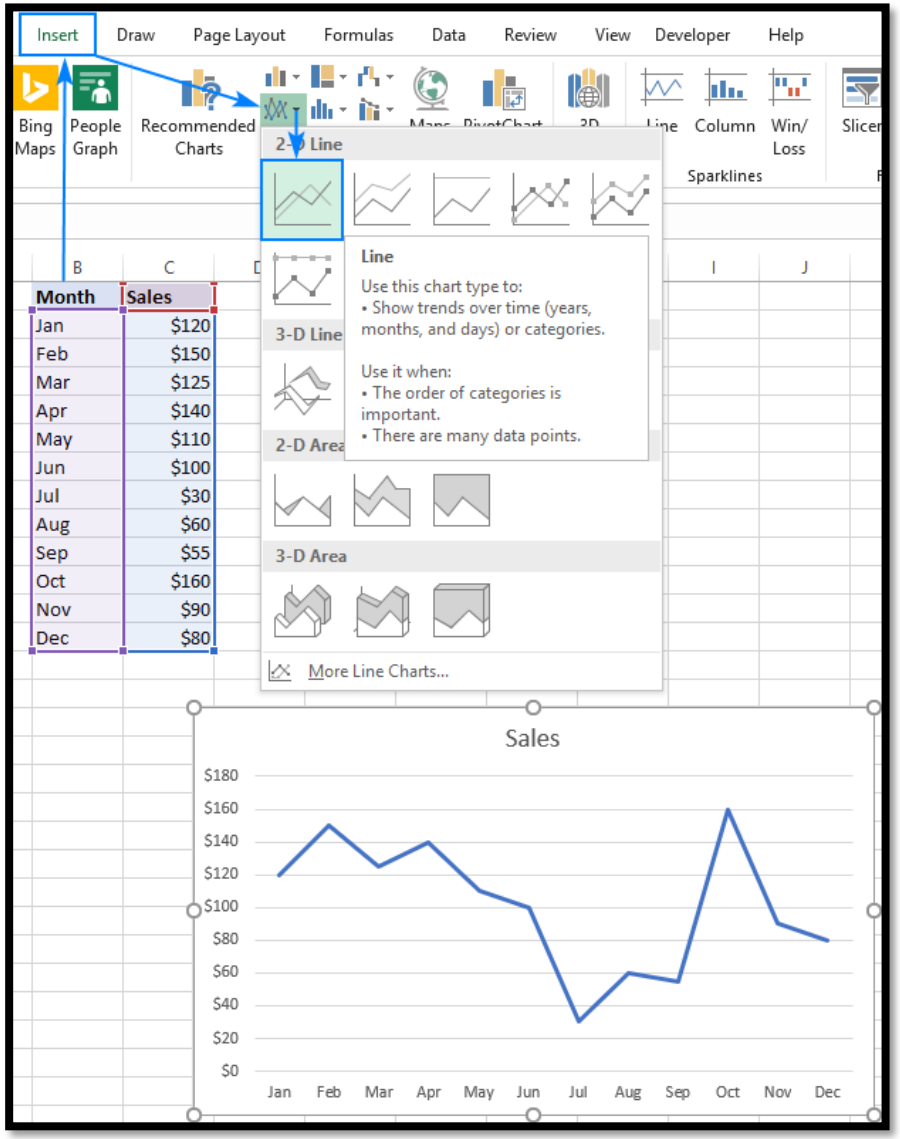


Creating a Line Chart

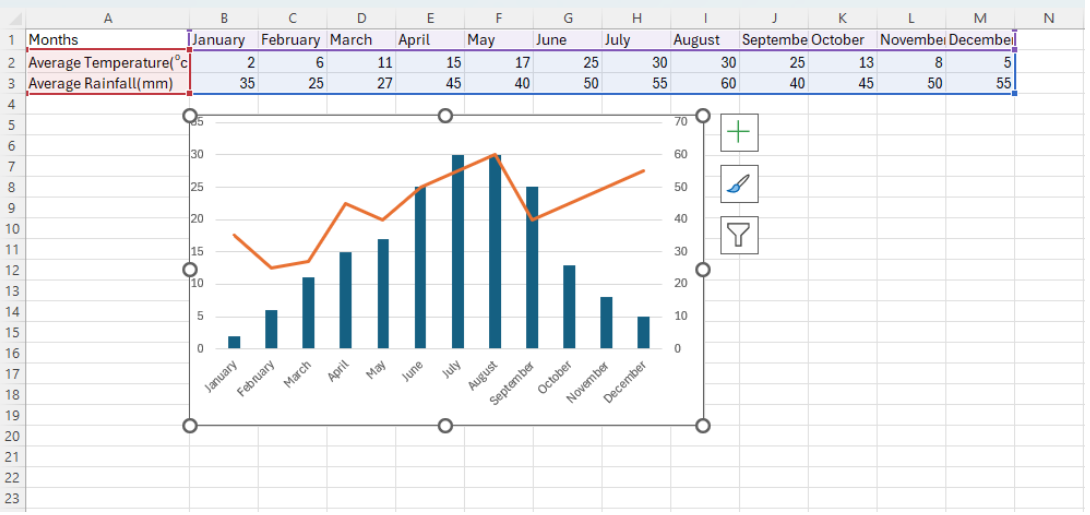
- *Great for showing progress over time.*
- *Use for time-series data (e.g., Days, Months).*

Steps:

1. *Insert → Chart → Line*
2. *Choose a format*
3. *Add titles and gridlines*



Advanced: The Combo Chart



Months	January	February	March	April	May	June	July	August	Septembe	October	Novembe	Decembe
Average Temperature(°c	2	6	11	15	17	25	30	30	25	13	8	5
Average Rainfall(mm)	35	25	27	45	40	50	55	60	40	45	50	55

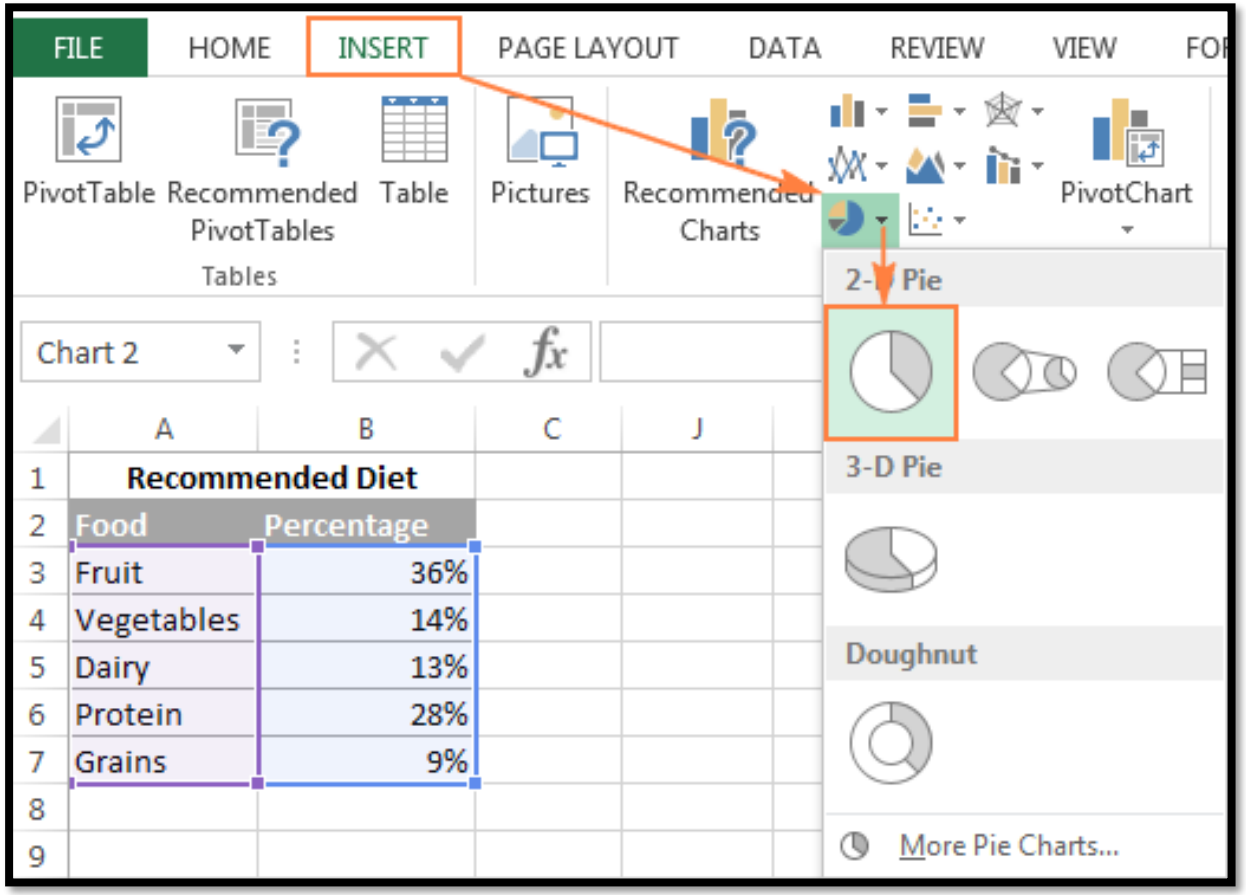
- The Problem: Plotting large numbers (Rs. 100,000) and small numbers (5%) together.
- The Solution: A Combo Chart with a Secondary Axis.
- How: Insert → Recommended Charts → All Charts → Combo.
- Crucial Step: Check the "Secondary Axis" box for the percentage data so it gets its own scale on the right side of the chart!

Creating a Pie Chart

- *Best for showing parts of a whole.*
- *Limited to 5–6 categories for clarity.*

Steps:

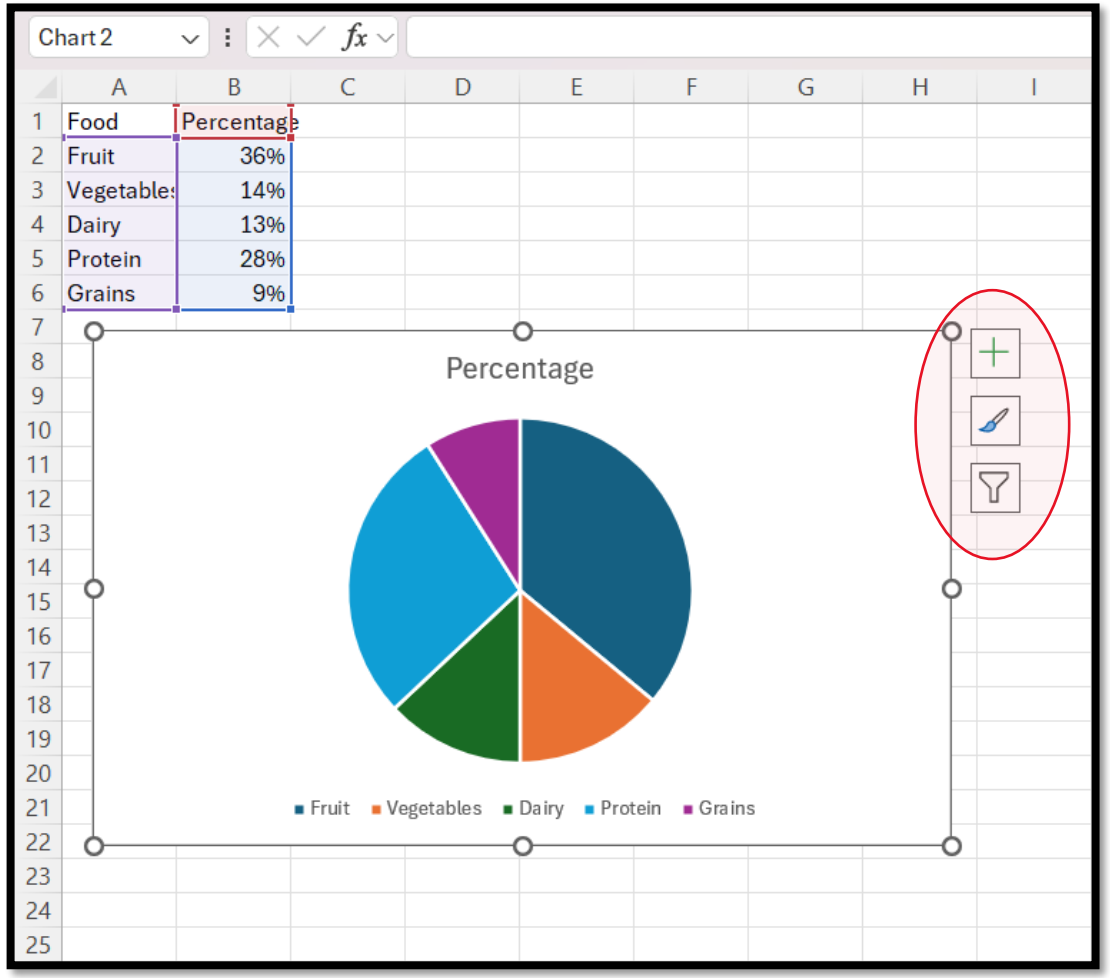
1. *Insert → Chart → Pie*
2. *Use data with totals*
3. *Add labels, colors, and percentages*



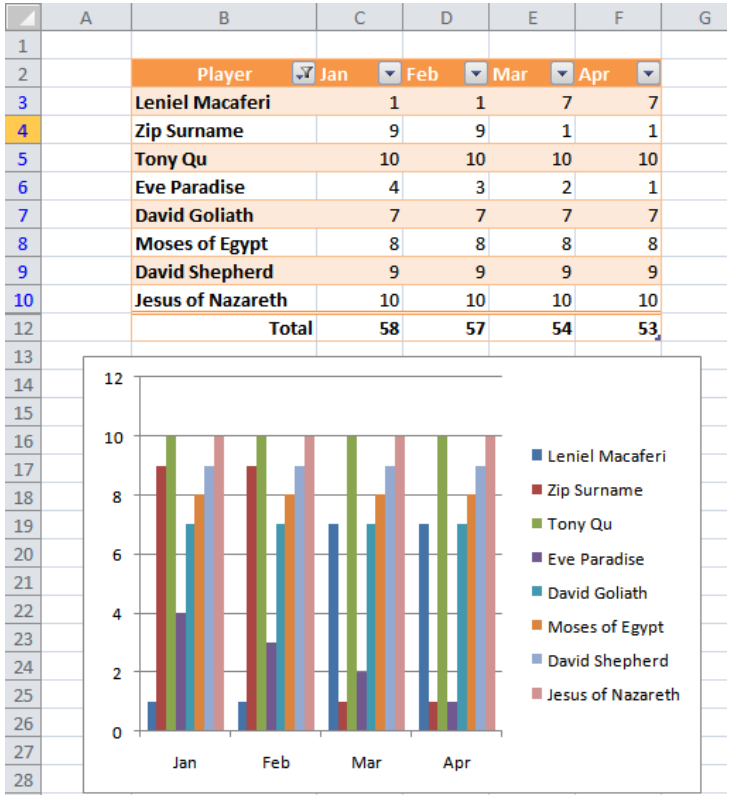
Customizing Charts

Click the chart → Chart Tools → Customize:

- *Title, Axis Labels*
- *Colors and Styles*
- *Gridlines and Legends*



Never Redraw a Chart Again (Dynamic Charts)



Player	Jan	Feb	Mar	Apr
Leniel Macaferi	1	1	7	7
Zip Surname	9	9	1	1
Tony Qu	10	10	10	10
Eve Paradise	4	3	2	1
David Goliath	7	7	7	7
Moses of Egypt	8	8	8	8
David Shepherd	9	9	9	9
Jesus of Nazareth	10	10	10	10
Total	58	57	54	53

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- Standard charts do not update when you type new data at the bottom.
- The Fix: Before you insert your chart, click your data and press Ctrl + T to format it as an official Excel Table.
- Now, insert your chart.
- The Magic: Type a new month or a new student at the bottom of your data.
- The chart will automatically expand and draw the new bar/line instantly!

Chart Design Tips

✓ *Use the right chart type*

✓ *Add labels and titles*

✓ *Avoid clutter*

✗ *Don’t overuse 3D or effects*

Practice Task

Open Excel and create:

1. *Bar Chart – Daily Study Hours*
2. *Line Chart – Monthly Mobile Data Usage*
3. *Pie Chart – Time Spent on Hobbies*

Bar Chart –
Daily Study Hours

Day	Study Hours
Monday	2
Tuesday	3
Wednesday	1.5
Thursday	2.5
Friday	3
Saturday	4
Sunday	2

Line Chart –
Monthly Mobile Data Usage (GB)

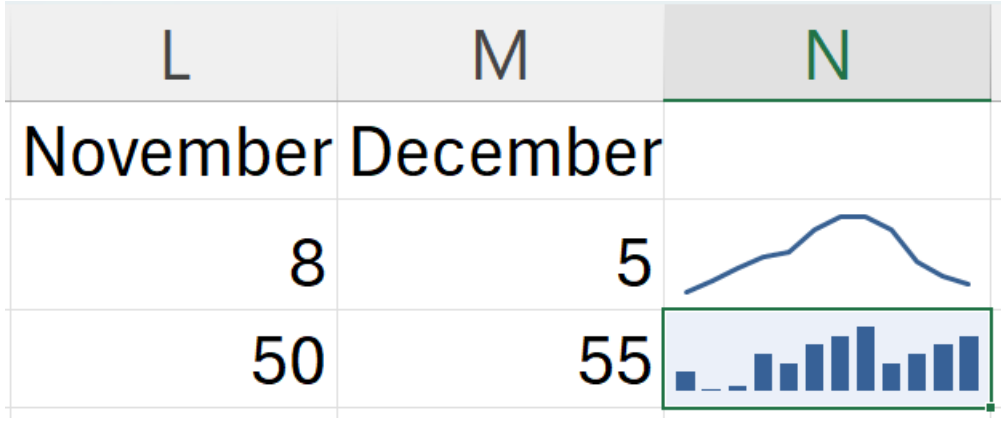
Month	Data Usage (GB)
January	12.5
February	10.2
March	13.0
April	11.4
May	14.1
June	12.8
July	15.6

Pie Chart –
Time Spent on Hobbies
(Hours per Week)

Hobby	Hours
Reading	5
Gaming	6
Music	4
Sports	3
Watching TV	2

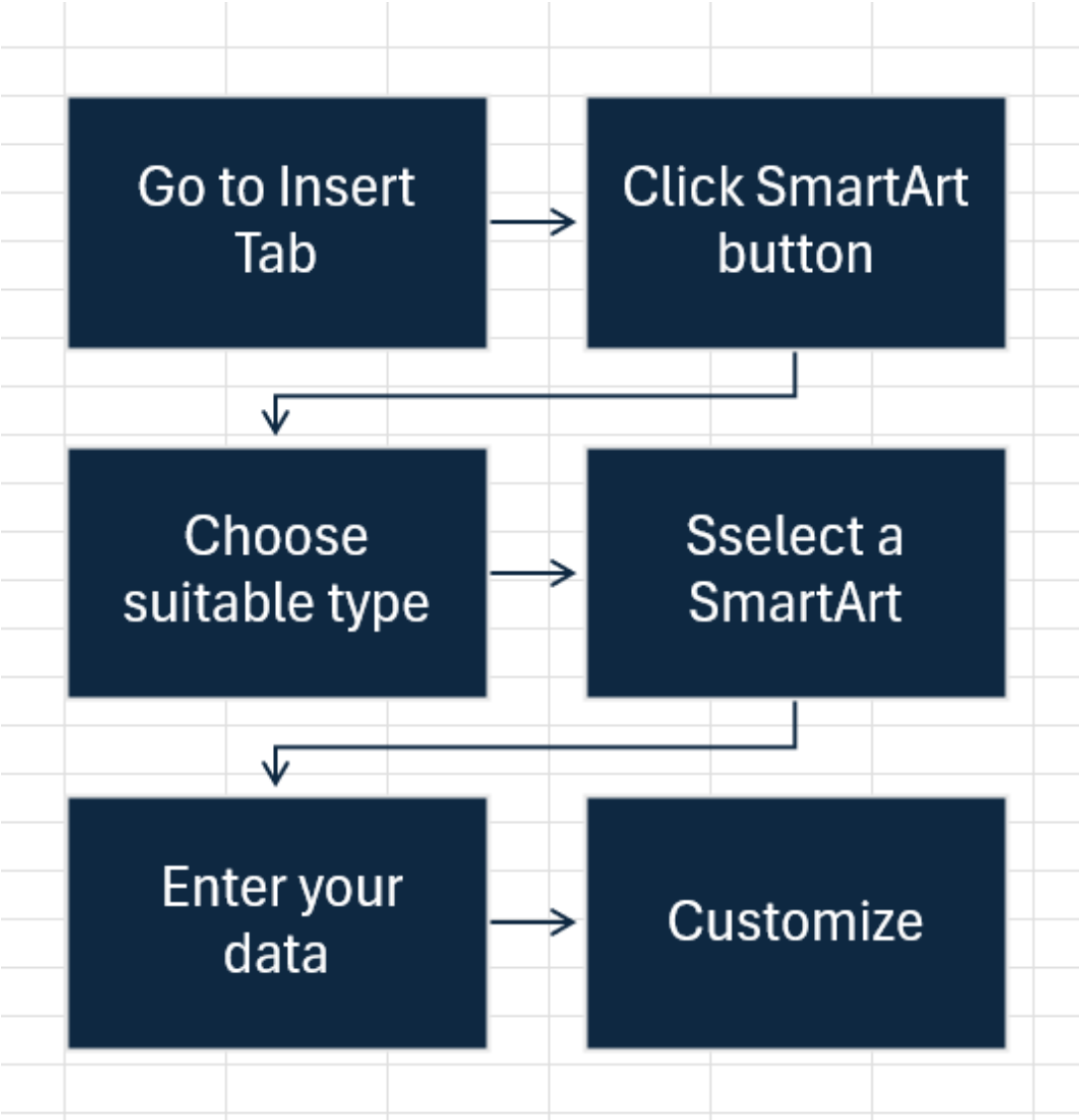
Pro Tip: Sparklines (Cell Charts)

- What if a chart takes up too much space?
- Sparklines are tiny, one-cell charts that show a trend perfectly.
- How: Click the empty cell at the end of a row of data.
- Go to Insert tab → Sparklines (Choose Line or Column).
- Select the data range for that row.
- Bonus: You can drag the fill handle down to instantly create mini-charts for everyone in the list!



SmartArt in Excel

- *Go to Insert → SmartArt*
- *Use for non-numerical visual explanation*
- *Example: “3 Steps to Learn Excel Charts”*
- *Create a SmartArt showing how to insert a SmartArt in Excel*



Recap & Questions

- *Bar, Line, Pie – each serves a purpose*
- *Visualizing data improves clarity*
- *Practice helps build skill!*

Next:

Excel Text & Logical Functions + Mini Project

Send Me Your Thoughts (Anonymously!)

Scan the QR code and share your questions or feedback

Thank You!

Student Name	English	Maths	Science	History
Nimal	78	82	85	70
Amal	65	88	74	72
Sunil	90	82	89	86
Kamal	70	75	91	75
Dilani	87	93	88	92